

Safety Information and Precautions

1.1 Safety Information

Please read this chapter carefully while installing and commissioning the inverter, and be sure to follow the safety precautions required in this chapter. We will assume no liability or responsibility for any injury or loss caused by improper operation.

In this manual, safety precautions are classified into the following two categories.



Danger

Indicates there is a risk of electric shock, which may cause equipment damage or personal injury if not avoided.



Warning

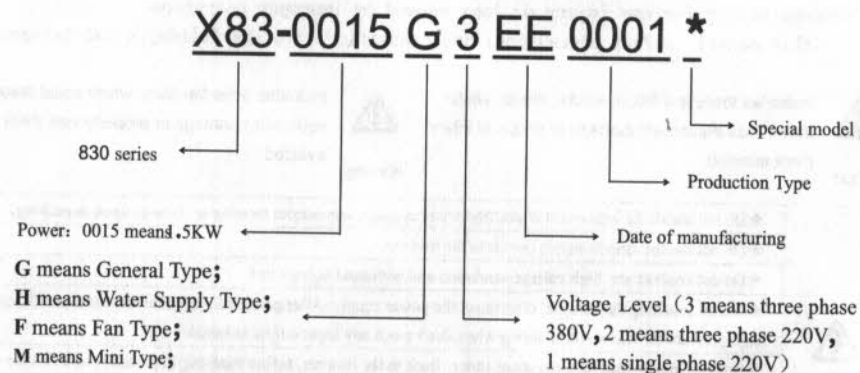
Indicates potential risks, which could result in equipment damage or property loss if not avoided

 Danger	★ Do not install the equipment if you find water seepage, component missing or damage upon unpacking!
	★ Do not use the strip to supply power to the inverter.
	★ Do not conduct any high voltage insulation and withstand voltage test.
	★ Before touching the inverter, disconnect the power supply; After power off, terminal and internal will exist high pressure for ten minutes, during when don't touch any input/output terminals.
	★ Rotating motor may feed electrical energy back to the inverter, before touching it, please ensure that the motor has stopped, or disconnected with the inverter.
	★ Before connecting the cable, make sure there is no voltage at the power terminal.
	★ Ground the inverter as standard. The ground wire must be able to withstand the maximum fault current limited by the fuse or circuit breaker.
 Warning	▲ Handle the equipment with care during transportation.
	▲ Keep away from combustibles and electrical conductors.
	▲ Inverters are best used indoors, IP20 inverters must be installed in a level 2 pollution environment or in the cabinet of the IP54 and higher level of protection.
	▲ Ensure adequate heat dissipation while installing the inverter and do not drill holes near it, for drilling dust and metal debris could fall into the inverter, which may leads to danger.
	▲ Do not drop wire end or screw into the inverter.
	▲ Never connect the power cables to the output terminals (U, V, W) of the inverter.
	▲ Never connect the braking resistor between the DC bus terminals DC+ and DC-.
	▲ Do not install any automatic control device between the inverter and the motor.
	▲ When the control cable is near the power line, keep a minimum spacing of 100 mm and arrange a 90-degree crossover. Make sure all the terminals have been fastened using the appropriate torque.
	▲ If the enable input signal is valid, the driven motor may start directly after being powered on.
	▲ Ensure that the supply voltage, frequency and phase are in accordance with the inverter rating.
	▲ When motor autotune, pay attention that the motor may rotate, which may cause danger.
	▲ The inverter can control the motor to run above or below the rated speed. When needing the motor to run over rated speed, you can confirm whether it is feasible with motor manufacturers.
	▲ Do not power on or off the inverter frequently, which may easy to shorten its service life. Please power on again ten minutes after power off.
	▲ In the area with an altitude of more than 1000m, derating is required.
	▲ Do not try to repair the inverter when errors and faults occur. Contact us for more help.

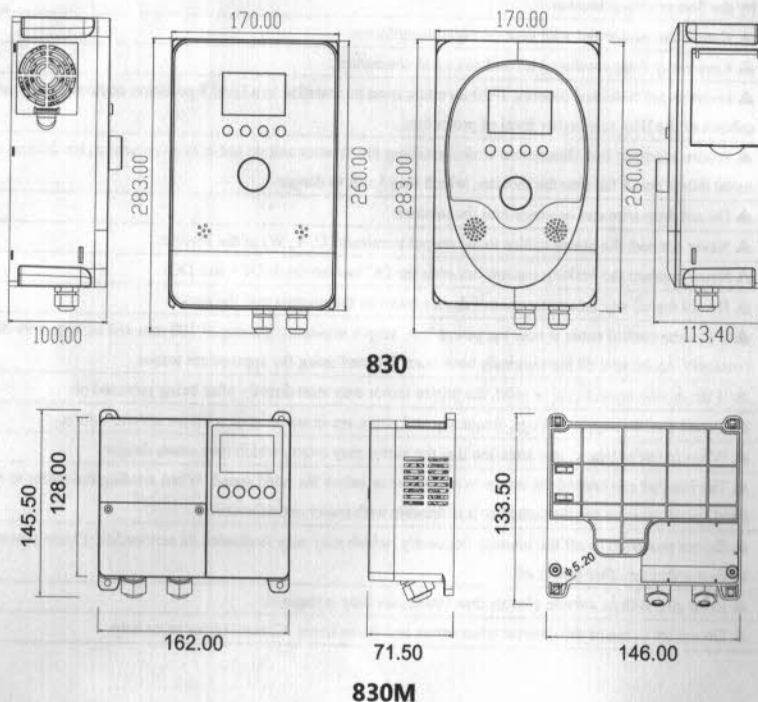
Product Information

2.1 Products Nameplate

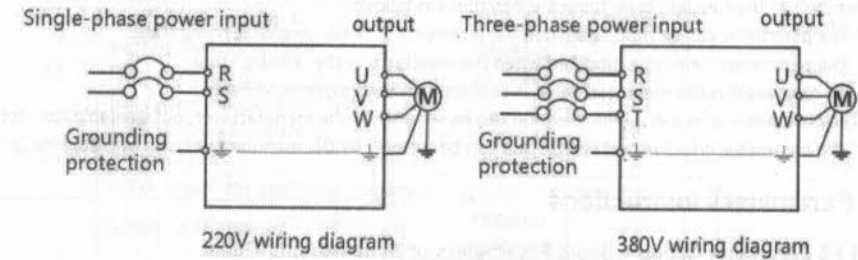
830 series inverters are named following rules below:



2.2 Dimensional drawing of product appearance



2.3 Basic wiring diagram:



2.4 Quick Commissioning Guide:

Asynchronous motor : 1. Set F3-00 =0 Asynchronous motor 2. Set F3-05 =0 3. Start

Synchronous motor :

1、The customer can provide the motor back EMF according to the following parameters settings.

Parameters	Description	Parameters	Description
F0-00	Motor rated power	F0-03	Motor rated frequency
F0-01	Motor rated voltage	F0-04	Motor rated speed
F0-02	Motor rated current	F0-05	Back EMF coefficient

After setting the parameters, press and hold the **ENTER** button for 3 seconds and the inverter will automatically go into static self-learning is finished, the inverter will start normally.

2、In the case that customer does not know the motor back EMF.

After setting the above parameters according to the motor parameters, the dynamic self-learning starts.

After setting F0-06=2, press the **ENTER** button, then enter the self-learning interface, press the **RUN** button to observe the motor status.

Dynamic self-learning when the motor first has an electromagnetic sound for a few seconds will rotate on its own, wait for the motor to stop automatically, the inverter will return to the speed interface, press the **RUN** button to start.

2.5 General Type Operation Panel

Operation Panel , which can modify the functional parameters of the inverter, monitor the working status of the inverter and control the operation of the inverter(start, stop), etc.

START STOP	Start up in keyboard operation mode Stop in keyboard operation mode.	▲	Increase the data or the function code.
PRGM ENT	Menu entry or exit Enter the menu to confirm the parameter settings. Long press for 3 seconds to enter the parameter setting.	▼	Decrease the data or the function code.

Parameters

The symbols in the function code table are described as follows:

"☆": The parameter can be modified when the inverter is in either stop or running state.

"★": The parameter cannot be modified when the inverter is in the running state.

"●": The parameter is the measured value in real-time and cannot be modified.

"*": The parameter is factory parameter and can be set only by the manufacturer, not available for user.

"▲": The parameter is factory parameter and can be set only by the manufacturer, not available for user.

3.1 Parameters Instructions

3.1.1 F0 Parameter Group – Basic Parameters of Synchronous Motor

Parameter	Description	Minimum Value	Default Value	Maximum Value	Unit	Permission
F0-00	Motor Rated Power	0.1	Depends on model	1000.0	kW	★
	This parameter is set to the rated power of the motor (nameplate).					
F0-01	Motor Rated Voltage	1	Depends on model	500	V	★
	This parameter is set to the rated voltage of the motor (nameplate).					
F0-02	Motor Rated Current	0.01	Depends on model	655.35	A	★
	This parameter is set to the rated current of the motor (nameplate).					
F0-03	Motor Rated Frequency	0	50.0	500.0	Hz	★
	This parameter is set to the rated frequency of the motor (nameplate).					
F0-04	Motor Rated Speed	1	Depends on model	65535	Rpm	★
	This parameter is set to the rated speed of the motor (nameplate).					
F0-05	Back EMF for PM Motor	0	Depends on model	6553.5	V	★
	This parameter is set as the back EMF of synchronous machine.					
F0-06	Motor Parameter Autotune	0	0	3	-	★
	0: No operation.					
	1: Static parameter identification. If the motor can't be completely separated from the load and can't rotate freely, please choose static parameter identification. 2: Dynamic parameter identification. If the motor is completely disconnected from the load and can rotate freely, please choose dynamic parameter identification. Note: After restoring the factory setting value, changing the model or setting the motor power and voltage level, it is necessary to identify the parameters again so that the vector control can run best.					
F0-07	Synchronous Motor Stator Resistance	0.001	Depends on model	65.535	Ω	★
F0-08	Synchronous Motor D-axis Inductance	0.01	Depends on model	655.35	mH	★
F0-09	Synchronous Motor Q-axis Inductance	0.01	Depends on model	655.35	mH	★
F0-07~F0-09 are asynchronous motor parameters, these parameters are generally not on						

	the motor nameplate, need to be obtained by motor parameter identification F0-06. If the induction motor cannot be tuned on site, you can input the above parameters according to the parameters provided by the motor manufacturer.					
F0-10	M-axis Current Loop Kp	0	2000	30000	-	☆
F0-11	M-axis Current Loop Ki	0	1000	30000	-	☆
F0-12	Starting Switching Frequency	1.0	Depends on model	F0-14	kHz	☆
	The carrier frequency at start.					
F0-13	SVC Low Speed Switching Frequency	0.8	Depends on model	F0-14	kHz	☆
	In SVC mode, the switching frequency of synchronous motor running at low speed.					
F0-14	Switching Frequency	1.0	Depends on model	16.0	kHz	☆
	This function is used to regulate the switching frequency of the inverter. When the switching frequency is low, the harmonic component of output current increases, the motor loss increases, and the temperature rise of the motor increases. When the switching frequency is high, the motor loss is reduced, the motor temperature decreases, but the inverter's loss increases, the inverter's temperature rise increases, and the interference increases.					
F0-15	Low Speed Maximum Magnetizing Current	0	30	80	%	☆
	Set the maximum excitation current of synchronous motor at low speed.					
F0-16	Torque Upper Limit	-200.0	150.0	+200.0	%	☆
F0-17	Deadtime Compensation	0	1	1	-	☆
	0: Disable 1: Enable					
F0-18	Voltage Feedback Enable	0	1	1	-	☆
	0: Disable 1: Enable					
F0-19	Command Source Selection	0	0	3	-	☆
	0: Panel control. Press the RUN key of the inverter to run, and press the STOP key to stop.					
	1: Terminal control. It is directly controlled by the inverter control terminal. By default, DI1 controls forward rotation and DI2 controls reverse rotation.					
F0-20	2: Communication control. It is controlled by Modbus RTU (RS485).					
	3: Auto start after power on.					
F0-21	Main Frequency Source Selection	0	1	9	-	★
	0: function code setting, power-off memory 1: panel potentiometer					
	2: AI1 3: AI2 (reserved)					
F0-22	4: Multi-segment command 5: PLC					
	6: Constant pressure water supply 7: General PID					
	8: Communication Settings 9: Reserved					
F0-21	Stop Mode	0	0	1	-	☆
	0: Coast to stop. After the shutdown command is effective, the inverter immediately stops the output, and the motor stops freely according to the mechanical inertia.					
	1: Ramp to stop. After the shutdown command is effective, the inverter reduces the output frequency according to the deceleration time, and stops after the frequency drops to 0.					
F0-22	Output Phase Sequence	0	0	1	-	☆
	0: U V W					
	1: U W V					

	Changing this parameter can change the motor direction without changing the motor wiring. Note: after the parameter is initialized, the parameter will return to the default value of 0, so be careful in some occasions where it is forbidden to change the motor direction.					
F0-23	Acceleration Time	0.1	Depends on model	500.0	second	☆
	The acceleration time required for the inverter to accelerate from 0 Hz to the upper limit frequency (F0-33).					
F0-24	Deceleration Time	0.1	Depends on model	500.0	second	☆
	The deceleration time required for the inverter to decelerate from the upper limit frequency (F0-33) to 0 Hz.					
F0-25	Synchronous Motor Initial Position Detection Mode	0	0	1	-	☆
	0: Check before each start. 1: No detection					
F0-26	Synchronous Motor Initial Position Identification Current Initial Value	5	70	180	%	★
F0-27	Main Menu Display Auto Switching	0	1	1	-	☆
	0: Switching is prohibited. When the display is switched from the frequency interface to other interfaces, it is forbidden to automatically switch back to the frequency interface. 1: Automatic switching. When the display is switched from the frequency interface to other interfaces, it will automatically switch back to the frequency interface after 10 seconds.					
F0-28	Parameter Modification Attribute	0	0	1	-	☆
	0: Allow modification. 1: No modification is allowed. When this parameter is set to 1, the inverter is forbidden to modify the parameter, and it must be set to 0 before it can be changed.					
F0-29	User Password	0	0	65535	-	★
	The inverter provides the user password protection function. When F0-29 is SET to non-zero, it is the user password. The password protection will take effect after exiting the function code editing state. Press the SET key again, "-----" will be displayed. You must input the user password correctly to enter the parameter interface.					
F0-30	Reserved					
F0-31	Reset to Factory Parameters	0	0	65535	-	★
	1: Reset the factory settings.					
F0-32	Load speed display coefficient	0.001	1.000	65.000	-	☆
F0-33	Frequency Upper limit	F0-34	50.0	500.0	Hz	☆
	Inverter maximum output frequency					
F0-34	Frequency Lower limit	0.0	0.0	F0-33	Hz	☆
	Inverter minimum output frequency					
F0-35	Stall Current Value	0	100	200	A	☆
F0-36	Stall Detection Time	0.0	3.0	3600.0	0.15	☆

F0-37	Current loop voltage feedforward mode	0	0	2	-	☆
F0-38	Autotune EMF Enable	0	0	1	-	☆
F0-39	Tune free mode	0	0	2	-	☆
F0-40	Control Frequency	2.0	4.0	16.0	Hz	☆
F0-41	Current Filter Coefficient	0	3	5		★
F0-42	Switching Frequency 2	F0-45	10.0	F0-33	Hz	☆
	When the running frequency is greater than this value, the speed loop PID parameter is selected as the speed loop parameter in the high-speed segment. Running frequency between high speed and low speed, the speed loop PID parameter linear transformation of two sets of PID parameters.					
F0-43	Proportional Gain 2	1	5	100	-	☆
	Setting the proportional coefficient of the speed regulator can adjust the speed dynamic response characteristics of vector control. Increasing the proportional gain can speed up the dynamic response of the speed loop, but excessive proportional gain may make the system oscillate. Note: The parameters of high-speed area and low-speed area are only valid when F3-00 selects vector control.					
F0-44	Integral Time Constant 2	0.01	1.00	10.00	second	☆
	The speed dynamic response characteristic of vector control can be adjusted by setting the integral time of the speed regulator. Shortening the integration time can accelerate the dynamic response of the speed loop, but too short integration time may make the system oscillate.					
F0-45	Switching Frequency 1	0.0	5.0	F0-42	Hz	☆
	When the operating frequency is less than this value, F0-46 and F0-47 are selected as PID parameters of speed loop.					
F0-46	Proportional Gain 1	1	30	100	-	☆
	The inverter runs at different frequencies and can select different speed loop PID parameters. When the running frequency is less than the switching frequency of the low speed segment F0-44, the proportional gain of the speed loop is used.					
F0-47	Integral Time Constant 1	0.01	0.50	10.00	second	☆
	When the operating frequency is less than the switching frequency F0-44 in the low speed section, the value of this parameter is used for the speed loop integration time.					
F0-48	Speed Loop Filter Time Constant	0	0.2	1.00	second	☆
	This parameter generally does not need to be adjusted, and the filtering time can be appropriately increased when the speed fluctuation is large. If the motor oscillates, the parameter should be appropriately reduced. The speed loop filter time constant is small, and the output torque of a inverter may fluctuate greatly, but the response speed is fast.					
F0-49	T-axis Current Loop Kp	0	2000	30000	-	☆
F0-50	T-axis Current Loop Ki	0	1000	30000	-	☆
F0-51	Low Speed Switching Frequency Switch Frequency	3.0	20.0	F0-33	Hz	☆
	At low speed, the switching frequency is the set value of F0-12. After running the set value of this parameter, the switching frequency changes to the set value of F0-14.					
F0-52	Low Speed Magnetizing Current Switching Frequency	3.0	20.0	F0-33	Hz	☆
	The maximum magnetizing current of synchronous motor at low speed is set at F0-15. After reaching this frequency, it will switch to normal current size.					

F0-53	Low Speed Magnetizing Current Switching Frequency Bandwidth	0.0	5.0	F0-33	Hz	☆
	When the synchronous motor runs at low speed, when the frequency reaches the set value of F0-52, if the current changes within the set range of F0-53, the low speed magnetizing current is switched only once.					
F0-54	SVC Initial Position Compensation Angle	0.0	0.0	359.9	°	☆
F0-55	Synchronous Motor Inductance Detection Current	30	80	120	%	☆
F0-56	Synchronous Motor Back EMF Identification Initial Current	0	50	180	%	★
F0-57	Synchronous Motor Back EMF Identification Final Current	30	80	180	%	★
F0-58	Synchronous Motor Tuning Current Loop Kp Adjustment Coefficient	1	6	100	-	☆
F0-59	Synchronous Motor Tuning Current Loop Ki Adjustment Coefficient	1	6	100	-	☆
F0-60	Synchronous Motor Flux Weakening Mode	0	1	2	-	☆
F0-61	0: Disable. The motor is not subject to flux-weakening control. At this time, the maximum speed of the motor is related to the bus voltage of the inverter. There is no flux-weakening current, and the output current is small, but the running frequency may not reach the set frequency. If you want to achieve a higher speed, you need to turn on the flux-weakening function. 1: Automatic adjustment. It is automatically adjusted by the inverter, and the higher the speed after entering the field weakening area, the greater the field weakening current. 2: Calculation + Automatic Adjustment. Combined with automatic adjustment, the speed of flux weakening current adjustment is faster, and this mode can be set when automatic adjustment can't meet the demand, but this mode depends on the accuracy of motor parameters.					
	Synchronous Motor Flux Weakening Coefficient	0	5	50	-	☆
F0-62	In the direct calculation mode, the required demagnetizing current can be calculated according to the target speed, and the size of demagnetizing current can be manually adjusted through F9-18. The smaller the demagnetizing current is, the smaller the total output current will be, but the desired flux weakening effect may not be achieved.					
	Flux Weakening Integral Multiple	2	2	10	-	☆
F0-63	Changing this parameter can change the adjustment speed of the flux weakening current. However, faster adjustment of the flux weakening current may lead to instability. Therefore, you do not need to manually change this parameter.					
	Saturation Margin for PM Motor	1	5	50	-	☆
F0-64	Maximum Torque Ratio Current Enable	0	0	1	-	☆
F0-65	0: Disable 1: Enabled					
	Salient Rate Gain Coefficient	50	100	500	-	☆
F0-66	Related to the structure of synchronous motor, according to the different characteristics of the motor to set different salient pole rate gain coefficient, generally no need to set.					
	Maximum output voltage coefficient	100	105	110	%	★

	The ability to increase the maximum output voltage of the inverter can be improved by increasing F0-66, which can increase the maximum load capacity of the weak magnetic zone of the fan. However, an increase in motor current ripple will increase the heat generation of the motor; On the contrary, the maximum load capacity of the motor in the weak magnetic zone will decrease, but the reduction of motor current ripple will reduce the heat generation of the motor. Generally, no adjustment is required.
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3.1.2 F2 Parameter Group – Terminal IO Function Selection

Parameter	Description	Minimum Value	Default Value	Maximum Value	Unit	Change Permission
F2-00	DI1 Terminal Function Selection	0	1	33	-	★
	0: No function 1: Forward running FWD 2: Reverse running REV 3: Fault reset 4: Coast stop, that is, blocking PWM output. 5: Running Pause 6: External fault input (normally open) 7: External fault normally closed input 8-33: Reserved					
F2-01	Relay Output Function Selection	0	1	7	-	☆
	The output terminal of each relay can provide 7 kinds of functions, these functions are: 0: No function. 1: The inverter is running. The inverter is in the running state, and when there is an output frequency (which can be zero), it outputs the ON signal. 2: Inverter fault. When the inverter fails and stops, it outputs the ON signal. 3: Ready for running. When the power supply of the main circuit and control circuit of the inverter is stable, and the inverter does not detect any fault information, and the inverter is in an operational state, the ON signal is output. 4: The upper limit frequency arrives. When the operating frequency reaches the upper limit frequency, the ON signal is output. 5: The lower limit frequency arrives. When the operating frequency reaches the lower limit frequency, the ON signal is output. This signal is OFF in the stop state. 6: Torque limit. In the speed control mode of the inverter, when the output torque reaches the torque limit, the inverter is in the stall protection state and outputs the ON signal at the same time. 7: Timer exceeded. When the running time of the inverter reaches the set timing time, it outputs the ON signal. 8: Press the programming and UP buttons together to switch the relay output status.					
F2-02	AI 1 Gain	0	1.00	20.00	-	★
	Analog input AI1 signal gain multiple, maximum gain up to 20 times. For example, AI1 is used as the target frequency setting, and this parameter is set to 2.00; Then a 5V input signal can make the Inverter run at the maximum frequency.					
F2-03	AI 1 Offset	-10.0	0	10.0	V	★
	Analog input 1 signal offset value, the maximum offset can be +/-10V. For example, AI1 is set as the target frequency and this parameter is set to 2.0; Then the 8V input signal can make the inverter run at the maximum frequency. Internal calculated value of AI1 = actual input * F2-02 + F2-03					
F2-04	Preset Frequency	0.0	50.0	F0-09	Hz	☆
	When the target frequency setting mode is selected as "Digital Setting", this parameter sets the initial value for the target frequency of the inverter. After the target frequency is modified by the "Up/Down" key, this parameter will become invalid temporarily, unless this parameter is modified again.					

F2-05	Low Frequency Action	0	0	2	-	☆
	0: Running at lower limit frequency 1: Stop 2: Zero-speed running When the set frequency is below the lower limit frequency, the running state of the inverter can be selected by this parameter.					
F2-06	Skip Frequency 1	0.0	0.0	F0-33	Hz	☆
	When the target frequency is set within the skip frequency range, the final running frequency of the inverter will avoid the range and run stably with the boundary value outside the range. It can be used to avoid the frequency resonance point of mechanical equipment. This parameter is the reference value of skip frequency, and the range is set by F2-07.					
F2-07	Skip Frequency Bandwidth	0.0	0.0	F0-33	Hz	☆
	Used in combination with F2-06, set specific skip frequency range (F2-06-F2-07) ~ (F2-06+F2-07). After this range is enabled, the actual operating frequency of the inverter is a hysteresis curve: when the frequency rises from low to within the range, the frequency remains at the low frequency boundary; When the frequency decreases from high to within the range, the frequency remains at the high frequency boundary;					
F2-08	LED2 Display Selection	0	2	10	-	☆
	0: Running Frequency 1: Running Speed 2: Output Current 3: DC Bus Voltage 4: Output Current 5: Output Power 6: AI1 Voltage 7: Motor Temperature Value 8: Heatsink Temperature Value 9: Actual Switching Frequency 10: Actual RPM					
F2-09	Set Inverter Power on Time	0	0	17520	hour	☆
	After the accumulated power-on time of the inverter exceeds this value, the inverter reports a fault Err20. The function of this parameter is invalid when it is set to 0.					
F2-10	Switching Frequency Adjusting with Temperature	0	1	1	-	☆
	When the inverter detects that the radiator temperature is high, it automatically reduces the switching frequency to reduce the temperature rise of the inverter. When the radiator temperature is low, the switching frequency gradually returns to the set value. This parameter is disabled when the value is set to 0.					
F2-11	Switching Frequency Adjusting Start Temperature	0	70	150	℃	☆
	When the inverter detects that the radiator temperature exceeds the set value of this parameter, F2-10 function is effective, and the switching frequency is adjusted with the temperature.					
F2-12	Switching Frequency Adjusting Time	0.1	20.0	50.0	s	☆

F2-13	DPWM Switching Upper Limit Frequency	5.0	50.0	F0-33	Hz	☆
	This parameter is valid only for VF control. When the asynchronous VF is running, the wave sending mode is 7-segment continuous modulation mode below this value, and on the contrary, it is 5-segment intermittent modulation mode. For 7-segment continuous modulation, the switch loss of a inverter is large, but the current ripple is small; The switching loss is small and the current ripple is large in the 5-segment discontinuous modulation mode. However, it may lead to instability of motor operation at high frequency, and generally does not need to be modified.					
F2-14	Fault Enable Selection 1	00000	01111	11111	-	☆
	0: Protection is prohibited. 1: Enable protection Bit: Motor overload protection selection Ten bit: Output open-phase protection. Hundred bit: Input open-phase protection. Thousand bit: Power-on short-circuit protection to ground. Ten thousand bits: output detection before operation (including grounding and phase loss)					
F2-15	Fault Auto Reset Times	0	0	20	time	☆
	Inverter can automatically reset after fault alarm. After this number is exceeded, the inverter will remain in a fault state. When set to 0, the automatic reset function is not enabled.					
F2-16	Fault Auto Reset Interval Time	0.1	1.0	100.0	second	☆
	The waiting time from the inverter fault alarm to the automatic fault reset.					
F2-17	Inverter Address	1	1	249	-	☆
	The local address when using the communication function of the inverter. When this value is set to 0, it is the broadcast address, which realizes the broadcast function of the upper computer.					
F2-18	Baud Rate	0	0	4	-	★
	0: 9600bps 1: 19200bps 2: 38400bps 3: 57600bps 4: 115200bps					
F2-19	Data Format	0	3	3	-	☆
	0: No check -2 stop bits (8-N-2) 1: Even check -1 stop bit (8-E-1) 2: Odd check -1 stop bit (8-O-1) 3: No check -1 stop bit (8-N-1)					
F2-20	Communication Timeout	0.0	0.0	60.0	second	☆
	When this parameter is set to 0.0 second, no communication timeout detection is performed. When this parameter is set to more than 0.1 second, if the interval between one communication and the next communication exceeds the communication timeout, the inverter will report a communication failure (Err16).					
F2-21	Reserved					
F2-22	Automatically Start Delay Time	0	150	3600	second	☆
F2-23	Cooling Fan Running Mode	0	0	1	-	☆
	0: The cooling fan runs when the temperature is higher than 45 C; 1: The cooling fan starts immediately when the inverter is running. When the inverter detects that the heat sink temperature exceeds the set value of F2-11, the switching frequency starts to adjust after the set time of F2-12.					

3.1.3 F3 Group- Asynchronous Motor Parameters

Parameters	Description	Minimum Value	Default Value	Maximum value	Unit	Change permission
F3-00	Motor control method	0	2	2	-	★
	0 : V/F control. 1: Vector speed control (IMSVC) of asynchronous motor. F0-06 parameter identification is required after SVC control is selected. 2: Vector speed control (FMSVC) of synchronous motor. F0-06 parameter identification is required after SVC control is selected.					
F3-01	Torque Boost	0	Depends on model	30.0	%	☆
	Under the V/F control mode, the output torque of the motor is relatively low in low frequency operation, which can increase the value of this parameter; However, the torque boost setting is too large, the motor is easy to overheat, and the inverter is easy to overcurrent. When the load is heavy and the starting torque of the motor is insufficient, it is recommended to increase this parameter. When the load is light, the torque can be reduced.					
F3-02	Torque Boost Cut-off Frequency	0.0	50.0	F0-33	Hz	★
	Below this frequency, the torque boost is effective, and beyond this set frequency, the torque boost fails.					
F3-03	VF Over-current Stall Action Current	50	150	200	%	★
F3-04	VF Overvoltage Stall Action Voltage	200.0	Depend on model	2000.0	V	★
	VF overvoltage stall running voltage.					
F3-05	Speed Tracking Start	0	1	1	-	☆
	0: Disable 1: Enable When the inverter starts, there is a short time delay to detect the motor speed, and control it from the current motor speed.					
F3-06	Speed tracking closed loop current Kp	0	Depends on model	1000	-	☆
	The F3-06 -F3-09 parameters do not need to be set by the user.					
F3-07	Speed tracking closed loop current Ki	0	Depends on model	1000	-	☆
F3-08	Speed tracking closed loop current value	30	Depends on model	200	%	☆
F3-09	Lower limit of speed tracking closed loop current	1.0	3.0	10.0	%	★
F3-10	Asynchronous motor stator resistance	0.001	Depends on model	65.535	Ω	★
F3-11	Rotor resistance of asynchronous motors	0.001	Depends on model	65.535	KW	★
F3-12	Asynchronous motor leakage inductance	0.01	Depends on model	655.35	mH	★

F3-13	Asynchronous motor mutual inductance resistance	0.1	Depends on model	6553.5	mH	★
F3-14	Asynchronous motor no-load current	0.01	Depends on model	F0-02	A	★
	F3-10~F3-14 are the parameters of the asynchronous motor, which are usually not available on the motor nameplate and need to be obtained by motor parameter identification F0-06. If the asynchronous motor cannot be turned on site, you can input the above parameters according to the parameters provided by the motor manufacturer.					

3.1.4 Monitoring Parameter

The monitoring parameters can only be read and cannot be modified.

Parameters	Description	Unit	Communication Address	Change permission
U0-00	Inverter Running State 1: forward 2: reverse 3: stop	-	1000H	▲
U0-01	Fault Code	-	1001H	▲
U0-02	Set Frequency	0.1Hz	1002H	▲
U0-03	Running Frequency	0.1Hz	1003H	▲
U0-04	Running Speed	Rpm	1004H	▲
U0-05	Output Voltage	V	1005H	▲
U0-06	Output Current	0.1A	1006H	▲
U0-07	Output Power	0.1KW	1007H	▲
U0-08	DC Bus Voltage	V	1008H	▲
U0-09	Output Torque	0.1Nm	1009H	▲
U0-10	Power Factor Angle	-	100AH	▲
U0-11	DI input state, default display 0, DI1_valid when set1	-	100BH	▲
U0-12	Reserved	-	100CH	▲
U0-13	AI1 Voltage Before Correction	0.01V	100DH	▲
U0-14	Reserved	-	100EH	▲
U0-15	AI1 Voltage	0.01V	100FH	▲
U0-16	Reserved	-	1010H	▲
U0-17	PID Setting	-	1011H	▲
U0-18	PID Feedback	-	1012H	▲
U0-19	Remaining Running Time	0.1Min	1013H	▲
U0-20	Current Power-on Time	Min	1014H	▲
U0-21	Current Running Time	0.1Min	1015H	▲
U0-22	Cumulative Running Time	Hour	1016H	▲
U0-23	Accumulated Power-on Time	Hour	1017H	▲
U0-24	Cumulative Power Consumption	Kwh	1018H	▲
U0-25	Motor Temperature Value	℃	1019H	▲
U0-26	IGBT Temperature Value	℃	101AH	▲
U0-27	Actual Switching Frequency	0.1KHz	101BH	▲
U0-28	M-axis Current Actual Value	0.1A	101CH	▲
U0-29	T-axis Current Actual Value	0.1A	101DH	▲
U0-30	Feedback Speed Actual Value	0.1Hz	101EH	▲
U0-31	Reserved	-	101FH	▲
U0-32	Reserved	-	1020H	▲
U0-33	Reserved	-	1021H	▲
U0-34	Reserved	-	1022H	▲
U0-35	Reserved	-	1023H	▲
U0-36	Reserved	-	1024H	▲
U0-37	Reserved	-	1025H	▲

U0-38	Reserved	-	1026H	▲
U0-39	Reserved	-	1027H	▲
U0-40	Reserved	-	1028H	▲
U0-41	Reserved	-	1029H	▲
U0-42	Product Serial Number Lower 16 Digits	-	102AH	▲
U0-43	Product Serial Number Higher 16 Digits	-	102BH	▲
U0-44	Motor Boot Version	-	102CH	▲
U0-45	CPU Type	-	102DH	▲
U0-46	Power Board Hardware Version	-	102EH	▲
U0-47	Power Board Software Version	-	102FH	▲
U0-48	Control Board Software Version	-	1030H	▲
U0-49	Product Number	-	1031H	▲
U0-50	Manufacturer Code	-	1032H	▲
U0-51	Third (most recent) Fault Code	-	1033H	▲
U0-52	Second Fault Code	-	1034H	▲
U0-53	First Fault Code	-	1035H	▲
U0-54	Third Fault Frequency	0.1Hz	1036H	▲
U0-55	Third Fault Current	0.1A	1037H	▲
U0-56	Third Fault DC Bus Voltage	0.1V	1038H	▲
U0-57	Third Fault Heatsink Temperature	℃	1039H	▲
U0-58	Third Fault Time(from power-on time)	Min	103AH	▲
U0-59	Third Fault Time(from running time)	0.1Hour	103BH	▲
U0-60	Second Fault Frequency	0.1Hz	103CH	▲
U0-61	Second Fault Current	0.1A	103DH	▲
U0-62	Second Fault DC Bus Voltage	0.1V	103EH	▲
U0-63	Second Fault Heatsink Temperature	℃	103FH	▲
U0-64	Second Fault Time(from power-on time)	Min	1040H	▲
U0-65	Second Fault Time(from running time)	0.1Hour	1041H	▲
U0-66	First Fault Frequency	0.1Hz	1042H	▲
U0-67	First Fault Current	0.1A	1043H	▲
U0-68	First Fault DC Bus Voltage	0.1V	1044H	▲
U0-69	First Fault Heatsink Temperature	℃	1045H	▲
U0-70	First Fault Time(from power-on time)	Min	1046H	▲
U0-71	First Fault Time(from running time)	0.1Hour	1047H	▲

Maintenance and Troubleshooting

4.1 Routine Maintenance

4.1.1 Regular Inspection

Due to the influence of environmental temperature, humidity, dust and vibration, the internal devices of the inverter will be aged, resulting in potential failures of the inverter or reducing the service life of the inverter. Therefore, it is necessary to carry out daily and regular maintenance of the inverter.

Daily Inspection Items	Regular Inspection Items
▲ Whether the sound of the motor changes abnormally or vibrates during running.	▲ Check whether the air channel is clean
▲ Does the installation environment of inverter change.	▲ Check whether the screws are loose.
▲ Whether the cooling fan of the inverter works normally and whether there are stains.	▲ Check whether the inverter is corroded.
▲ Is the inverter overheated.	▲ Check the wiring terminals for traces of arcing pulling.
▲ Is the inverter kept clean.	

4.1.2 Long-time Storage

If the inverter has been stored for a period of time before installation, or has not been powered by the main power supply for a long time, it is necessary to age and energize the DC capacitor in the inverter according to the following instructions before operation, and the inverter can run normally after the aging is completed.

Storage Time	Input Voltage 1	Duration 1	Input Voltage 2	Duration 2	Input Voltage 3	Duration 3	Input Voltage 4	Duration 4
Less than 1 year	100%							
1-2 years	100%	1 hour						
2~3 years	25%	0.5 hour	50%	0.5 hour	75%	0.5 hour	100%	0.5 hour
More than 3 years	25%	2 hours	50%	2 hours	75%	2 hours	100%	2 hours

Without treatment

Normal running

4.2 Faults and Solutions

If the inverter system fails during operation, the inverter will stop output immediately to protect the motor. At the same time, the inverter fault relay acts. The inverter panel displays fault codes. The following table lists the fault types and common solutions corresponding to the fault codes.

The list in the table is for reference only. Do not repair or modify it without authorization. If you can't troubleshoot, please ask the supplier for technical support.

Fault Name	Display	Possible Causes	Solutions
Inverter Unit Protection	Err01	- The output circuit is grounded or short circuited - The connecting cable of the motor is too long - The module overheats - The internal connections become loose - The main control board is faulty - The drive board is faulty - The inverter module is faulty	- Eliminate external faults - Install a reactor or an output filter fan - Check the air filter and the cooling fan - Connect all cables properly - Contact for technical support - Contact for technical support - Contact for technical support
Overcurrent During Acceleration	Err02	- The output circuit is grounded or short circuited - The control method is vector and no parameter identification - The acceleration time is too short - Manual torque boost or V/F curve is not appropriate - The voltage is too low - The startup operation is performed on the rotating motor. - A sudden load is added during acceleration - The inverter model is of too small power class	- Eliminate external faults - Perform the motor auto-tuning - Increase the acceleration time - Adjust the manual torque boost or V/F curve - Adjust the voltage to normal range - Select rotational speed tracking - Restart or start the motor after it stops - Remove the added load. - Select higher power rating inverter
Overcurrent During Deceleration	Err03	- The output circuit is grounded or short circuited - The control method is vector and no parameter identification - The deceleration time is too short - The voltage is too low - A sudden load is added during deceleration - The braking unit and braking resistor are not installed	- Eliminate external faults - Perform the motor auto-tuning - Increase the deceleration time - Adjust the voltage to normal range - Remove the added load. - Install the braking unit and braking resistor
Overcurrent at Constant Speed	Err04	- The output circuit is grounded or short circuited - The control method is vector and no parameter identification - The voltage is too low - A sudden load is added during deceleration - The inverter model is of too small power class	- Eliminate external faults - Perform the motor auto-tuning - Adjust the voltage to normal range - Remove the added load. - Select higher power rating inverter

Fault Name	Display	Possible Causes	Solutions
Overvoltage During Acceleration	Err05	- The input voltage is too high - An external force drives the motor during acceleration - The acceleration time is too short - The braking unit and braking resistor are not installed	- Adjust the voltage to normal range - Cancel the external force or install a braking resistor - Increase the acceleration time - Install the braking unit and braking resistor
Overvoltage During Deceleration	Err06	- The input voltage is too high - An external force drives the motor during deceleration - The deceleration time is too short - The braking unit and braking resistor are not installed	- Adjust the voltage to normal range - Cancel the external force or install a braking resistor - Increase the deceleration time - Install the braking unit and braking resistor
Overvoltage at Constant Speed	Err07	- The input voltage is too high - An external force drives the motor during running	- Adjust the voltage to normal range - Cancel the external force or install a braking resistor
Control Power Supply Fault	Err08	The input voltage is not within the allowable range	Adjust the voltage to normal range
Undervoltage	Err09	- Instantaneous power failure - The inverter's input voltage is not within the allowable range - The DC bus voltage is abnormal - The rectifier bridge and buffer resistor are faulty - The drive board is faulty - The main control board is faulty	- Reset the fault - Adjust the voltage to normal range - Contact for Technical support - Contact for Technical support - Contact for Technical support - Contact for Technical support
Inverter Overload	Err10	- The load is too heavy or locked rotor occurs on the motor - The inverter model is of too small power class	- Reduce the load and check the motor and mechanical condition - Select an inverter of higher power class
Motor Overload	Err11	- The load is too heavy or locked rotor occurs on the motor - The inverter model is of too small power class	- Reduce the load and check the motor and mechanical condition - Select higher power rating inverter
Power Input Phase Loss	Err12	- The three-phase power input is abnormal - The drive board is faulty - The lightning board is faulty - The main control board is faulty	- Eliminate external faults - Contact for Technical support - Contact for Technical support - Contact for Technical support
Power Output Phase Loss	Err13	- The cable connecting the inverter and the motor is faulty - The inverter's three-phase outputs are unbalanced when the motor is running - The drive board is faulty - The module is faulty	- Eliminate external faults - Check whether the motor three-phase winding is normal - Contact for Technical support - Contact for Technical support
Module Overheat	Err14	- The ambient temperature is too high - The air filter is blocked - The fan is damaged - The thermally sensitive resistor of the module is damaged - The inverter module is damaged	- Lower the ambient temperature - Clean the air filter - Replace the damaged fan - Replace the damaged thermally sensitive resistor - Replace the inverter module

4.3 Common Faults and Solutions

The following faults may be encountered during the use of the inverter. Refer to the following table for simple fault analysis:

SN	Fault	Possible Causes	Solutions
1	There is no display at power-on	1. There is no power supply to the inverter or the power input to the inverter is too low 2. The power supply of the switch on the drive board of the inverter is faulty 3. The rectifier bridge is damaged 4. The buffer resistor is faulty 5. The control board or the operation panel is faulty 6. The cable connecting the control board and the drive board and the operation panel breaks	1. Check the power supply 2. Check the DC bus voltage 3. Re-connect the cables 4~6. Contact us for technical support
2	"Err23" is displayed at power-on	1. The motor or the motor output cable is short circuited to the ground 2. The inverter is damaged	1. Measure the insulation of the motor and the output cable with a megger 2. Contact us for technical support
3	Err14 (Module overheat) fault alarm frequently	1. The setting of switching frequency is too high 2. The cooling fan is damaged or the air filter is blocked 3. Components inside the inverter are damaged (thermocouple or other)	1. Reduce the switching frequency (F0-14) 2. Replace the fan and clean the air filter 3. Contact us for technical support
4	The motor does not rotate after the inverter runs	1. Check the motor and the motor cables 2. The inverter parameters are set improperly (Motor parameters) 3. The cable between the drive board and the control board is in poor contact 4. The drive board is faulty	1. Ensure the cable between the inverter and the motor is normal 2. Replace the motor or clear mechanical faults 3. Check the re-set motor parameters 4. Contact us for technical support
5	The DI terminals are disabled	1. The parameters are set incorrectly 2. The external signal is incorrect 3. The control board is faulty	1. Check and reset the parameters in group F2 2. Re-connect the external signal cables 3. Contact us for technical support
6	The inverter overcurrent and overvoltage frequently	1. The motor parameters are set improperly 2. The acceleration/deceleration time is improper 3. The load fluctuates	1. Re-set motor parameters or re-perform the motor auto-tuning 2. Set proper acceleration/deceleration time 3. Contact us for technical support

Fault Name	Display	Possible Causes	Solutions
External Equipment Fault	Err15	1. External fault signal is input via DI 2. External fault signal is input via virtual I/O	1. Reset the operation 2. Reset the operation
Communication Fault	Err16	1. The controller is in abnormal state 2. The communication cable is faulty 3. Parameters setting incorrect	1. Check the cabling of host computer 2. Check the communication cabling 3. Check communication parameters
Contact Fault	Err17	1. The drive board and power supply are faulty 2. The contactors is faulty	1. Replace the faulty drive board or power supply board 2. Replace the faulty contactor
Current Detection Fault	Err18	1. The HALL device is faulty 2. The drive board is faulty	1. Replace the faulty HALL device 2. Replace the faulty drive board
Motor Auto-tuning Fault	Err19	1. The motor parameters are not set according to the nameplate 2. The motor auto-tuning times out	1. Set the motor parameters according to the nameplate properly 2. Check the cable connecting the inverter and the motor
EEPROM Write Fault	Err21	1. The EEPROM chip is damaged	1. Replace the main control board
Inverter Hardware Fault	Err22	1. Overvoltage 2. Overcurrent	1. Solve as overvoltage fault 2. Solve as overcurrent fault
Short Circuit to Ground	Err23	1. The motor is short circuited to the ground	1. Replace the cable or motor
Accumulative Running Time Reached	Err26	1. The accumulative running time reaches the setting value	1. Clear the record through the parameter initialization function
Accumulative Power-on Time Reached	Err29	1. The accumulative power-on time reaches the setting value	1. Clear the record through the parameter initialization function
Pulse-by-pulse Current Limit Fault	Err40	1. The load is too heavy or locked rotor occurs on the motor 2. The inverter model is of too small power class	1. Reduce the load and check the motor and mechanical condition 2. Select an inverter of higher power class
Motor Switchover Fault During Running	Err41	1. Change the selection of the motor via terminal during running of the inverter	1. Perform motor switchover after the inverter stops
Excessive Speed Deviation Fault	Err42	1. Excessive Speed deviation - Inspection parameter setting is not correct 2. No parameter identification	1. Correctly Setting Parameter 2. Executive parameter identification
Water Shortage alarm	A52	1. Pressure sensor is damaged 2. Check whether the parameters of the inverter are incorrectly set 3. Whether the pipe network and motor are correct	1. Check pressure sensor 2. Check inverter parameter setting 3. Check motor and pipe
Overpressure Fault	Err53	1. Pressure sensor is damaged 2. Check whether the parameters of the inverter are incorrectly set	1. check the pressure sensor 2. Test whether the inverter is correctly set

4.4 Warranty Agreement

Free warranty only refers to the inverter itself.

In case of failure or damage within the warranty terms, our company is responsible for 12 months warranty (from the date of delivery, the bar code on the fuselage shall prevail, and if there is a contract agreement, it shall be implemented according to the agreement). For more than 12 months, we will charge a reasonable maintenance fee;

During the warranty period, our company will charge a certain maintenance fee if:

Machine damage caused by the user's failure to comply with the regulations in the user manual;

Machine damage caused by mistakes in use and unauthorized maintenance and modification;

Damage caused by fire, flood, abnormal voltage, etc.;

Damage caused by using the inverter for abnormal functions;

Damage caused by man-made falling and transportation after purchase;

Failure and damage caused by obstacles outside the machine (such as external equipment factors);

The service fee is calculated according to the uniform standard of the manufacturer. If there is a contract, the contract will take precedence.

If there is any problem in the service process, please contact the supplier in time.

The final interpretation right of warranty instructions belongs to our company.

Customer Information	Address:	
	Name:	Contact:
	Postal code:	Tel:
Product Information	Product model:	
	Fuselage Bar code (posted here):	
Fault Information	Agent name:	

Certificate	
This product has gone through rigorous quality control tests at factory.	
Inspector	
Approval Mark	